THE COMPOSITION CONSTRUCTION COMMUTES WITH PROPER PUSHFORWARD

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Push-pull compatibility of the bivariant action (relative-cycles chapter, Lemma lemma-action-push-pull, canonical tag 0H6B). cited at proof lines 2917-2918. Supplies the fibrewise identity f_*(g^* alpha cap -) = alpha cap g_*(-) underlying the family statement.

Source reference. Original lemma 0H6B: sources/stacks-chow/tags.txt, source lines 21416–21416. Cited at proof lines 2917-2918: the fibrewise identity.

Lemma 0.2. *Let S be a scheme. Let*

$$\begin{array}{ccc} X' & \xrightarrow{f} & X \\ \downarrow & & \downarrow \\ Y' & \xrightarrow{g} & Y \end{array}$$

be a cartesian diagram of schemes locally of finite type over S with g proper. Let $r, e \geq 0$. Let α be a family of r-cycles on the fibres of X/Y. Let β' be a family of e-cycles on the fibres of Y'/S. Then we have $f_*(g^*(\alpha) \circ \beta') = \alpha \circ g_*\beta'$.

Proof. Unwinding the definitions, this follows from Lemma 0H6B. □

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